

Understanding the spatial distribution of population in Copperbelt province, Zambia

Data provided:

- ***ZMB_adm2_gadm41_Copperbelt.shp***: Administrative boundaries (Level 2) for Copperbelt province, Zambia from GADM version 4.1, in Shapefile format
- ***zmb_ppp_2020_constrained.tif***: Gridded (raster) dataset of population estimates (WorldPop Global 2020 constrained estimates for Zambia), in GeoTiff format
- ***GHS_SMOD_E2020_GLOBE_R2023A_54009_1000_V1_0_R11_C21.tif***: Gridded (raster) dataset of rural/urban classes (GHS-SMOD, tile ID: R11_C21), in GeoTiff format

GHS-SMOD L2 classification		GHS-SMOD L1 classification	
Code	Class	Code	Class
10	Water		
11	Very low density rural	1	Rural
12	Low density rural		
13	Rural		
21	Suburban or peri-urban	2	Urban cluster
22	Semi-dense urban		
23	Dense urban		
30	Urban centre	3	Urban centre

Table 1: GHS-SMOD L1 and L2 classification

In this task, your aim is to gain an understanding of the spatial distribution of population in Copperbelt province, Zambia, including identifying which administrative units have the highest and lowest counts of population and the distribution of population across rural and urban contexts. You will be working with geospatial datasets from openly available sources.

1. Start by exploring the provided datasets, remembering to check the metadata (e.g. the spatial resolution and coordinate/projection system information). You may want to make notes of your observations.

I started by loading the three layers and checking their metadata and basic integrity before doing any processing. The Copperbelt ADM2 boundaries contain 12 polygons in EPSG:4326 with the expected ID and name fields (for example, GID_2, NAME_2). The WorldPop 2020 constrained population raster is a single-band Float32 GeoTIFF in EPSG:4326 on a 0.0008333 degree grid. The GHS-SMOD settlement tile is a single-band Int16 categorical raster in World Mollweide (ESRI:54009) on a 1 km grid.

Dataset	Format	CRS	Units	resolution	Data type	No Data
zmb_ppp_2020_constrained.tif (WorldPop population)	GeoTIFF	EPSG:4326 (WGS84)	degrees	3 arc-seconds	Float32	-99999
GHS_SMOD_..._R11_C21.tif (GHS-SMOD L2)	GeoTIFF	ESRI:54009 World Mollweide	metres	1000 m × 1000 m	Int16	-200
ZMB_adm2_gadm41_Copperbelt.shp (ADM2 boundaries)	Shapefile	EPSG:4326 WGS84	degrees	n/a	Mult Polygon	n/a

Table 2: Input datasets and technical specifications.

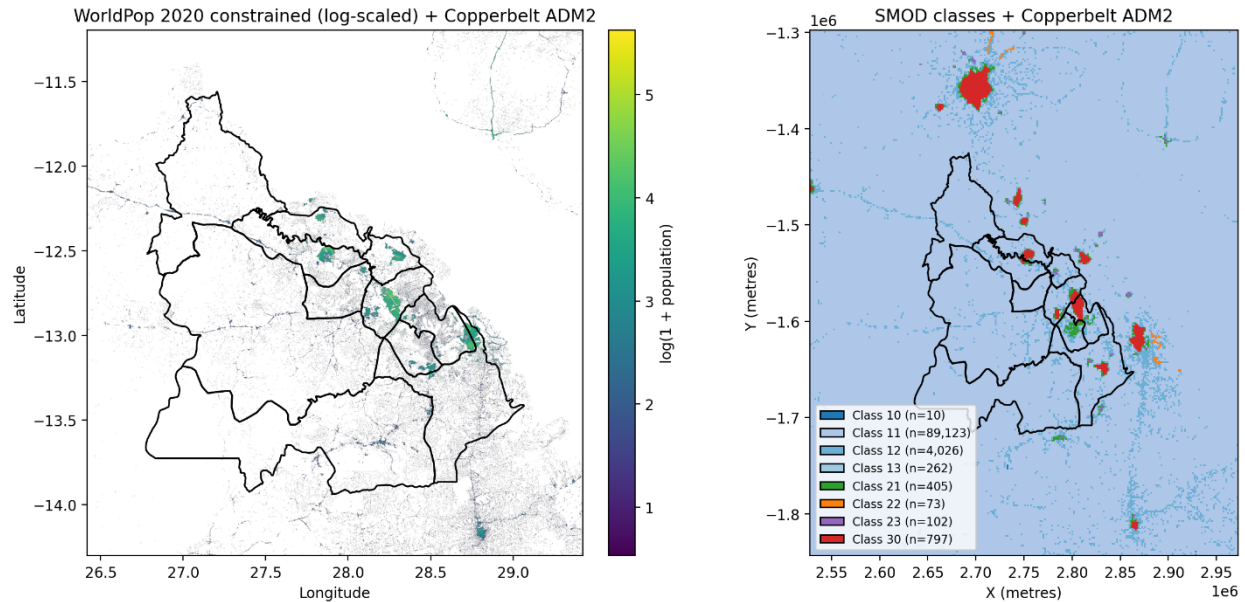


Fig 1. spatial baseline: WorldPop population surface (left) and SMOD categorical settlement classes (right)

2. The administrative boundaries have 2 errors in the dataset – can you identify these? Think about how you might address these issues if you were to encounter them in your work.

Q. What are the 2 errors in the administrative boundary dataset?

One polygon is duplicated as an exact geometry copy, and one feature is out of scope for Copperbelt (Mushindano). I confirmed the out-of-scope feature via the ADM1 attribute (NAME_1) and I confirmed duplication via a geometry fingerprint. For this exercise I kept the file ‘as provided’ where instructed, but I documented the issues.

3. As we're interested in understanding the spatial distribution of population, calculate the total population per administrative unit. There is no need to correct the administrative boundary to answer the following question. Please use the boundary as is.

Q. Which administrative unit (name) has the highest and lowest population count?

I calculated total population per district using a zonal sum of the WorldPop 2020 constrained population raster within the provided ADM2 polygons. I kept the boundary layer as supplied, as instructed, and computed a pop_sum field for each unit. I then ranked districts by pop_sum to identify the maximum and minimum. Based on the as-provided boundaries, the highest total was Kitwe and the lowest was Mushindano. If I restrict to Copperbelt districts only, the lowest becomes Kalulushi.

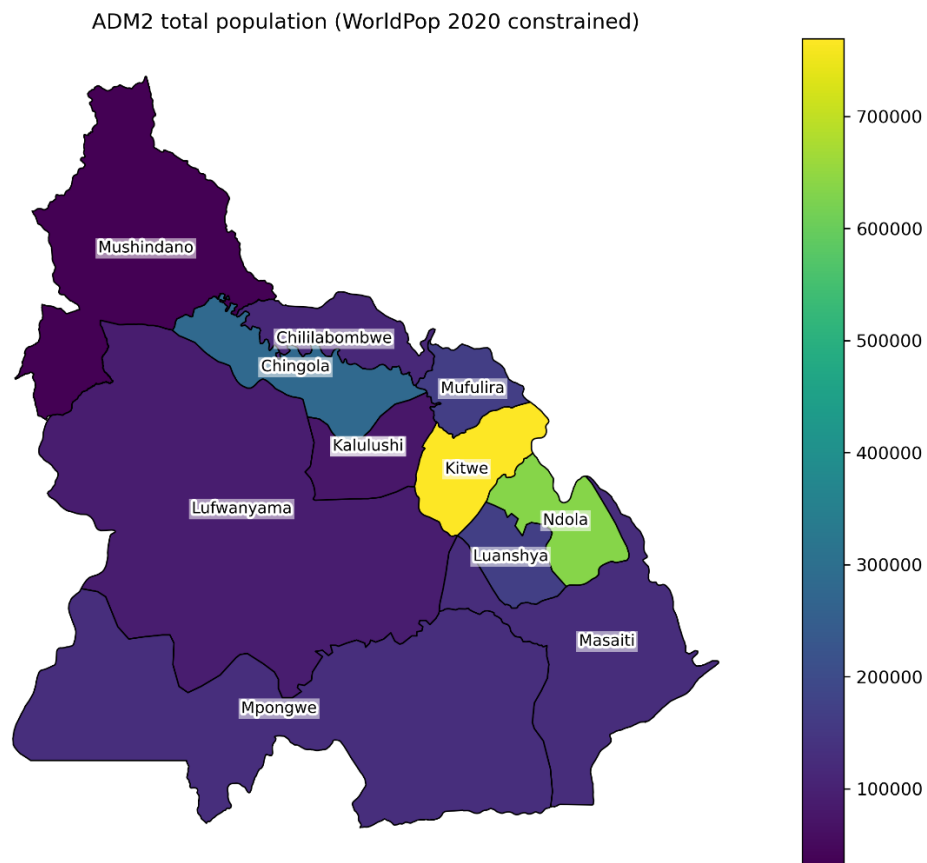


Fig 2. Copperbelt ADM2 population totals from WorldPop 2020 constrained raster

4. Copperbelt province includes dense urban and very rural areas. The GHS-SMOD classification provides a stratification of rural/urban classes. The L1 classification includes 3 classes, whilst the L2 classification provides a more granular stratification of 7 classes (table 1). In the data provided, you will find a gridded dataset for the L2 classification – if you haven't already, explore this data now.

Q. What data processing is needed to create a gridded version of the SMOD L1 classification (3 classes)? If you want to use the SMOD raster with the population raster, what additional processing steps are needed to ensure the data is harmonised?

I started from the GHS-SMOD Level-2 categorical raster and created a three-class Level-1 grid by reclassifying the L2 codes into broader rural–urban categories. In QGIS I did this in a single reproducible step using the GDAL raster calculator, writing an Int16 output so the classes remain clean and categorical, and assigning all non-target values to NoData so water and undefined areas are excluded from later summaries.

If SMOD is then used alongside the WorldPop population raster, the additional requirement is spatial harmonisation: the rasters must share a common CRS and a common pixel grid (resolution and alignment). Because SMOD is categorical, any reprojection/resampling must use nearest neighbour. The choice of reference grid is driven by the question: area-dominance is best handled on an equal-area SMOD grid, while population-by-class is typically safest by aligning SMOD to the population grid to avoid resampling the population surface.

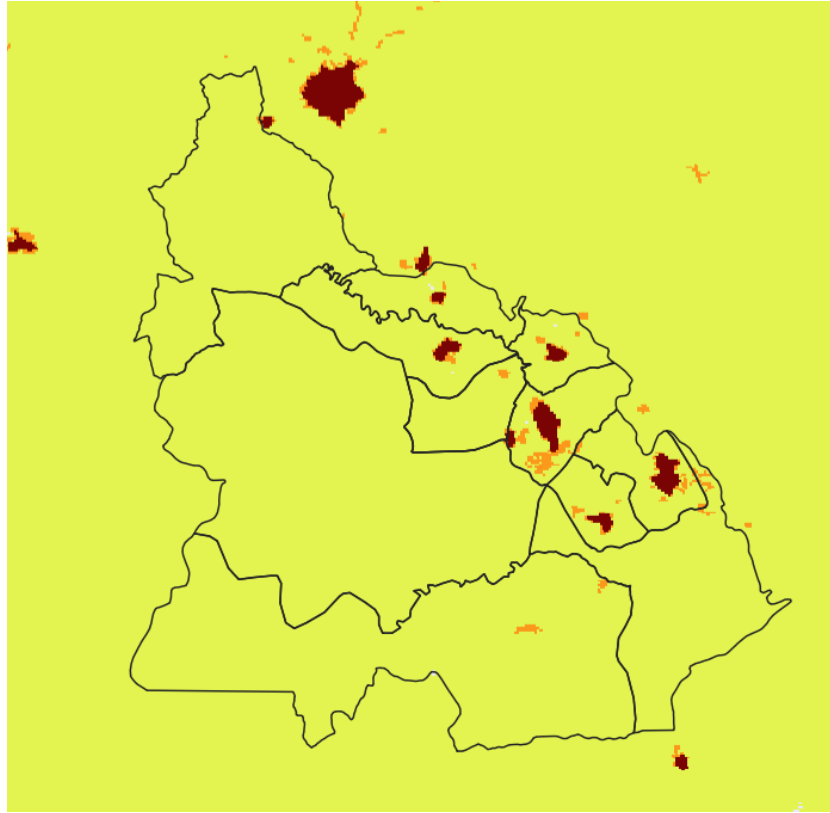


Fig 3. GHS-SMOD Level-1 settlement classes (3-class recode) over Copperbelt ADM2. The original SMOD L2 grid was recoded to L1 and displayed as three settlement types: Rural (C1) shown in yellow, Urban cluster (C2) shown in orange, and Urban centre (C3) shown in dark red.

5. We are interested in understanding the proportion of the total population in each rural/urban type. As the L2 classification is very granular, we will use the 3-class L1 classification (as detailed in table 1). Firstly, identify the dominant GHS-SMOD L1 class for each administrative unit in terms of geographic area (hint: there is no need to calculate areas, the number of grid cells is sufficient).

Q. For how many administrative units is “urban centre” the dominant SMOD-class (considering the geographic area or number of grid cells)? How many administrative units are predominantly “rural”?

For the dominant class by area, I first put the district boundaries into the same CRS as the SMOD grid, then I counted how many L1 grid cells of each class fall inside each district using zonal histograms and labelled the district by whichever class has the highest cell count. On that basis, none of the districts are dominated by “urban centre”, and all of the districts are predominantly rural by area.

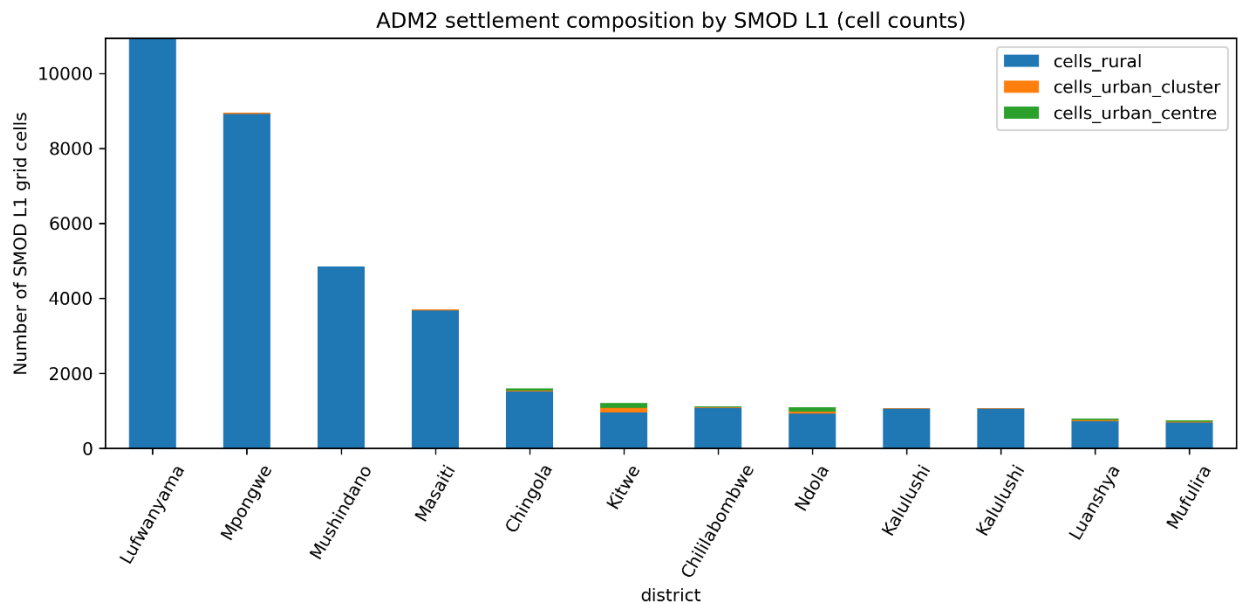


Fig 4. ADM2 settlement composition by SMOD L1 (cell counts).

6. Secondly, calculate the population in each GHS-SMOD L1 class within each administrative unit.

Q. For each SMOD L1 class, which administrative unit had the greatest proportion of population in each class? Which had the least?

To quantify population by settlement type within each ADM2 district, I first harmonised the categorical SMOD L1 raster to the WorldPop population grid so the two rasters were co-registered for pixel-wise analysis. In practice, that meant matching CRS (EPSG:4326), pixel size, grid origin, extent, and NoData handling, using nearest-neighbour resampling for SMOD to preserve class labels. With the grids aligned, I created three class-specific population surfaces by masking the WorldPop raster using SMOD L1 (C1 rural, C2 urban cluster, C3 urban centre), retaining population values where the class matched and assigning zero elsewhere so only the target class contributes to the sum. I then ran zonal statistics for each class to obtain per-district class totals, converted these into within-district proportions, and finally identified the districts with the highest and lowest proportions for each class.

Using those proportions:

- Class C1 (Rural):
 - Lowest proportion: Kitwe
 - Highest proportion: Lufwanyama
- Class C2 (Urban cluster):
 - Lowest proportion: Lufwanyama
 - Highest proportion: Kalulushi
- Class C3 (Urban centre):
 - Lowest proportion: Masaiti and Mpongwe (both 0, i.e., no population fell inside C3 within those districts)
 - Highest proportion: Chingola

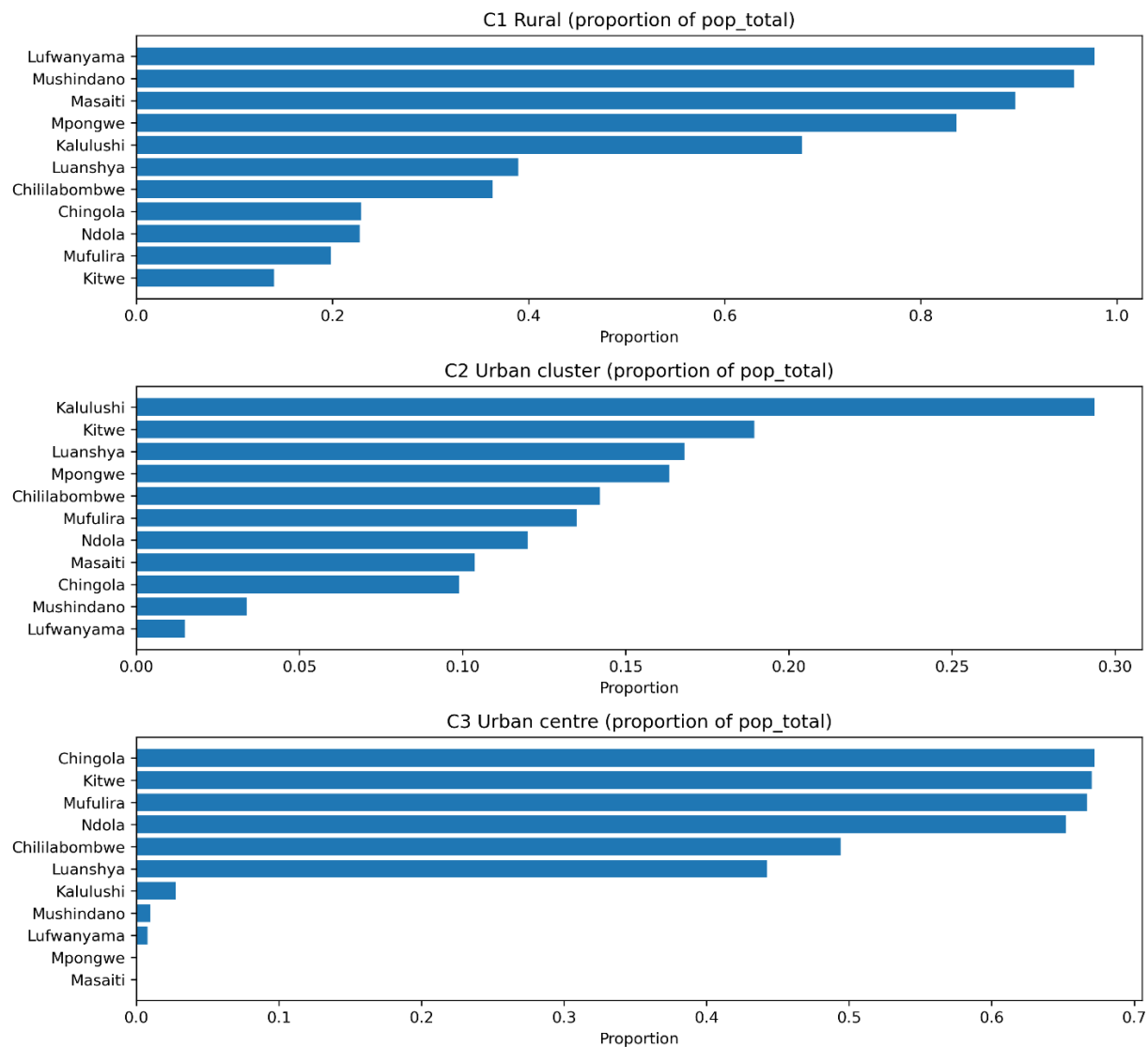


Fig 5. ADM2 population composition by SMOD L1 (ranked proportions)

Appendix — Reproducible Python workflow (outputs, QA, and requirements)

A small, reproducible Python workflow was used to validate inputs, generate district-level summaries, and produce reporting-ready figures. Outputs are written into three folders with consistent task prefixes (`task1_`, `task2_`, ...).

- *outputs/*
 - QA and reproducibility artefacts (`*.json`) recording inputs, parameters, CRS/grid metadata, and checks.
 - Geospatial outputs created during processing (e.g., SMOD aligned to the WorldPop grid as GeoTIFF).
- *tables/*
 - District-level results (`*.csv`), including:
 - Total population per ADM2 (zonal sums)
 - SMOD L1 composition per ADM2 (counts/proportions)
 - Per-class extremes tables (min/max with tie-handling)
- *figures/*
 - Saved plots (`*.png`), including:
 - Choropleth maps (totals or proportions by ADM2)
 - Ranked bar charts (district comparisons)
 - Stacked composition plots
 - Small-multiple maps (C1/C2/C3 proportions)

Python requirements

geopandas, rasterio, matplotlib, pandas, numpy

Code availability:

GitHub: <https://github.com/am636/copperbelt-worldpop-smod.git>